



The orbital dynamics modeling and simulation of asteroid probe based on variational integrators

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Abstract

Near-Earth Asteroids impact earth threatening human beings in the long term, hence, exploring asteroids and their complex surrounding dynamical environment is of both strategic importance and practical urgency. This work carries out orbital dynamics modeling and simulation for asteroid probes based on the inverse problem of Lagrangian mechanics and discrete Hamilton's principle: firstly, constructing the orbital dynamic model of asteroid probes by combining the law of universal gravitation and Newton's second law and solving the Lagrangian function by using the inverse problem of Lagrangian; secondly, establishing the discretized orbital dynamic model for asteroid probes based on the discrete Hamilton's principle; finally, performing numerical simulation on the situations with and without the gravitational perturbations of asteroid based on the variational integrators. The results have shown that the variational integrators can provide accurate real-time simulation of orbits and energy in the long-term.

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1. Introduction

Celestial bodies that orbit the Sun, range in size from approximately 1 m to 800 km, and do not actively release gas or dust are generally classified as asteroids. To date, a total of 38,481 Near-Earth Asteroids has been identified and monitored. Research on asteroids is of critical importance for understanding the early formation and evolution of the solar system and the origin and development of life on Earth, while simultaneously highlighting the potential existential threat they pose to humanity's survival and

advancement (Ouyang, 2014; Nuth et al., 2018). Consequently, the exploration, development, and utilization of asteroids, together with the formulation of effective strategies to mitigate the risks associated with potential impacts, are not only urgent practical imperatives but also matters of profound and far-reaching strategic significance (Zhang et al., 2024). At the same time, these endeavors present significant scientific and technological challenges to the global astronomical and aerospace communities, as well as to related disciplines (Xin, 2017; Wu et al., 2022; Gong et al., 2020). All such scientific and exploratory missions fundamentally rely on analyses of asteroid dynamic environments and in-depth investigations of probe orbital dynamics (Zeng et al., 2021). The central challenge in Near-Earth Asteroid research lies in the effective design and optimization of probe orbital trajectories (Cui and

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Qiao, 2013; Yu and BaoYin, 2014; Scheeres, 2016; Wang et al., 2022; Cui et al., 2022). The irregular morphologies and rotational states of asteroids, combined with the uncertainty of their operational environments, impose stringent requirements on the optimal orbit design and control of asteroid probes (Zeng and Wen, 2022). Accurate modeling and long-term simulation of such complex dynamical systems rely heavily on numerical integration methods. Traditional numerical integration methods, such as the Runge-Kutta method, have been extensively applied to the numerical solution of dynamical systems due to their simplicity and computational efficiency (Faleichik and Moisa, 2025). However, these methods do not preserve the intrinsic geometric structures of mechanical systems—such as energy, momentum, or symplecticity—which often leads to numerical dissipation or artificial energy drift in long-term simulations. Therefore, to achieve higher stability and structure-preserving performance, researchers have increasingly turned to geometric approaches in recent years.

Geometric mechanics employs modern differential geometry tools—including manifolds, fiber bundles, Lie groups, and Lie algebras—to formulate dynamical equations in a coordinate-independent and globally consistent form. This approach not only highlights the geometric structure of dynamical systems but also clarifies their physical implications (Niu et al., 2025). As a result, geometric mechanics has found wide applications across various disciplines (Thompson, 1987; Arnold et al., 2006; Marsden and Ratiu, 1999; Guo et al., 2007). Building on geometric mechanics, Marsden developed the variational integrators for mechanical systems based on the discrete Hamilton's principle, providing an effective approach for long-term, high-precision simulations of dynamical systems (Marsden and West, 2001). Bloch et al. integrated the variational integrators with geometric control theory and nonholonomic systems, thereby developing a long-term, high-precision control algorithm that maintains the stability of nonholonomic control systems (Bloch et al., 2003). By applying the geometric analogy method and incorporating the moving frame theory, Shi et al. constructed variational integrators for the Hamel field (Shi et al., 2017; Shi et al., 2020; Zenkov et al., 2012). Geometric mechanics and geometric numerical integration provide rational design principles and high-precision algorithms for efficient, real-time control systems, including spacecraft, submersibles, and robots. The integration of symmetric reduction with geometric numerical integration enables high-accuracy results while simplifying complex dynamical systems. Recent studies have applied these methods to nonholonomic mechanical systems (Xiong, 2022) and flexible multibody systems (Chen et al., 2022; Chen et al., 2023). Nevertheless, research on applying geometric numerical integration to orbital dynamics modeling and numerical simulation of asteroid probes remains limited (Koon et al., 2000; Moor et al., 2012; Bolatti and de Ruiter, 2021).

In response to the requirements of asteroid exploration missions, this research employs analytical mechanics to develop a relative orbital dynamics model between the probe and the asteroid. Subsequently, variational integrators are used to simulate the relative motion of the two bodies. Comparison with results from the traditional Runge-Kutta algorithm demonstrates that variational integrators offer higher precision and efficiency for such dynamical problems. Consequently, this study provides a high-precision numerical method for the optimal design of asteroid probe orbits.

2. Variational integrators

Hamilton's principle for constrained mechanical systems can be formulated as:

$$\delta S = \delta \int_{t_1}^{t_2} L(q, \dot{q}, t) dt = 0 \quad (1)$$

Here $L(q, \dot{q}, t)$ is the Lagrangian function that characterizes the dynamical system, where q is a generalized coordinate, $\dot{q}$ is the generalized velocity. The quantity $S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$ is referred to as the action function. Given the Lagrangian function L , the differential equation of motion for the dynamical system can be derived from Hamilton's principle (1). Instead of considering a position q and a velocity $\dot{q}$, we now take a time-step h and two points q_0 and q_1 lying on a curve at time h apart, such that $q_0 = q(0)$ and $q_1 = q(h)$, then we obtained a discrete Lagrangian $L_d(q_0, q_1, h)$. Next consider a discrete curve of points $\{q_k\}_{k=0}^N$. The action sum is obtained by summing the discrete Lagrangian on each adjacent pair. Assuming that boundary points q_0 and q_N are held fixed, this gives:

$$\delta S_d = \delta \sum_{k=0}^{N-1} L_d(q_k, q_{k+1}, h) = 0 \quad (2)$$

By using the endpoint condition $\delta q_0 = \delta q_N = 0$, the discrete Euler-Lagrange equation can be derived (Marsden and West, 2001):

$$D_1 L_d(q_k, q_{k+1}, h) + D_2 L_d(q_{k-1}, q_k, h) = 0, \quad (k = 1, 2, \dots, N-1) \quad (3)$$

Here $D_i (i = 1, 2)$ denotes the differentiation with respect to the i -th variable. And the relation must hold for each i . By applying the Legendre transform, the dynamics originally formulated on the tangent bundles TQ are mapped to the cotangent bundles T^*Q , thereby reformulating the system from the Lagrangian system to the Hamiltonian system. Now, we define the discrete Legendre transform which maps the discrete state space $Q \times Q$ to T^*Q , they can be written as follows:

$$\begin{aligned} \mathbb{F}^+ L_d : (q_0, q_1) &\mapsto (q_1, p_1) = (q_1, D_2 L_d(q_0, q_1)) \\ \mathbb{F}^- L_d : (q_0, q_1) &\mapsto (q_0, p_0) = (q_0, -D_1 L_d(q_0, q_1)) \end{aligned} \quad (4)$$

Hence, the discrete Hamilton equations can be derived:

$$p_k = -D_1 L_d(q_k, q_{k+1}, h) \quad p_{k+1} = D_2 L_d(q_k, q_{k+1}, h) \quad (5)$$

When the system is subjected to additional external forces-such as friction, external control forces, dissipative forces, and mechanical loads-the motion curve of the system satisfied the d'Alembert-Lagrange principle (Marsden and West, 2001):

$$\delta \int_{t_1}^{t_2} L(q, \dot{q}, t) dt + \int_{t_1}^{t_2} f_L(q, \dot{q}, t) \cdot \delta q dt = 0 \quad (6)$$

Here, f_L denotes a fiber-preserving mapping from TQ to T^*Q , referred to as the Lagrange force. In coordinate form, it can be expressed as follows:

$$f_L : (q, \dot{q}, t) \mapsto (q, f_L(q, \dot{q}), t)$$

The discrete Lagrange force f_L induces a mapping $f_d^\pm$: $Q \times Q$ to T^*Q , which can express in coordinate form as follows:

$$\begin{aligned} f_d^+(q_k, q_{k+1}, h) &= (q_{k+1}, f_d^+(q_k, q_{k+1}), h) \\ f_d^-(q_k, q_{k+1}, h) &= (q_k, f_d^-(q_k, q_{k+1}), h) \end{aligned} \quad (7)$$

When the discrete Lagrangian function L_d and the discrete Lagrange force $f_d^\pm$ are incorporated into the d'Alembert-Lagrange principle (6), the discrete d'Alembert-Lagrange principle can be written as:

$$\begin{aligned} &\delta \sum_{k=0}^{N-1} L_d(q_k, q_{k+1}, h) \\ &+ \sum_{k=0}^{N-1} [f_d^-(q_k, q_{k+1}, h) \cdot \delta q_k + f_d^+(q_k, q_{k+1}, h) \cdot \delta q_{k+1}] = 0 \end{aligned} \quad (8)$$

The discrete Euler-Lagrange equation in the presence of external forces can be derived by applying the endpoint conditions:

$$\begin{aligned} &D_1 L_d(q_k, q_{k+1}, h) + D_2 L_d(q_{k-1}, q_k) + f_d^+(q_{k-1}, q_k, h) \\ &+ f_d^-(q_k, q_{k+1}, h) = 0 \end{aligned} \quad (9)$$

3. Dynamic modeling of asteroid probe based on the inverse problem of Lagrange mechanics

3.1. Inverse problem of Lagrange mechanics

The inverse problem of Lagrangian mechanics, also referred to as the inverse problem of the variational method (Ding, 2016), involves determining the Lagrangian function of a system. This is achieved on the premise that the fundamental form of the second-order differential equation (system) is known. If the equations of motion for a local, holonomic, non-conservative mechanical system with n degrees of freedom are given as follows (Santilli, 1978):

$$\ddot{q}_i - \rho_{ij}(t, q) \dot{q}_j - \sigma_i(t, q) = 0, \quad (i, j = 1, 2, \dots, n) \quad (10)$$

where $\rho_{ij} = -\rho_{ji}$ represents an antisymmetric tensor. When Eq. (10) satisfies the following self-adjoint condition:

$$\begin{aligned} \rho_{ij} + \rho_{ji} &= 0 \\ \frac{\partial \rho_{ij}}{\partial q^k} + \frac{\partial \rho_{jk}}{\partial q^i} + \frac{\partial \rho_{ki}}{\partial q^j} &= 0 \quad (k = 1, 2, \dots, n) \\ \frac{\partial \rho_{ij}}{\partial t} &= \frac{\partial \sigma_i}{\partial q^j} - \frac{\partial \sigma_j}{\partial q^i} \end{aligned} \quad (11)$$

Consequently, the Lagrangian function of the mechanical system (10) can be expressed as:

$$L = \frac{1}{2} \dot{q}^i \dot{q}_i + D_i(t, q) \dot{q}^i + C(t, q) \quad (i = 1, 2, \dots, n) \quad (12)$$

The variables D_i and C satisfy the following partial differential equations:

$$\begin{cases} \frac{\partial D_i}{\partial q^j} - \frac{\partial D_j}{\partial q^i} = -\rho_{ij} \\ \frac{\partial C}{\partial q^i} = \frac{\partial D_i}{\partial t} - \sigma_i \end{cases} \quad (13)$$

By replacing the variables $(t, \tau q)$ with (t, q) , and restricting the integral domain to the star domain, the integration of Eq. (13), yields the following expressions for D_i and C :

$$\begin{cases} D_i = -\left\{ \int_0^1 [\tau \cdot \rho_{ij}(t, \tau q) d\tau] \right\} q^j \\ C = \left[\int_0^1 \sigma_i(t, \tau q) d\tau \right] q^i \end{cases} \quad (i, j = 1, 2, \dots, n) \quad (14)$$

Substituting Eq. (14) into Eq. (12), the Lagrangian function of the differential equation of motion (10) that satisfies the self-adjoint condition can then be obtained (Rubens, 2018).

3.2. Dynamic modeling of asteroid probe

Consider the three-body problem consisting of the Sun, an asteroid moving in an elliptical orbit around it, and a probe operating near the asteroid. Since both the asteroid and the probe are subjected by the Sun-centered gravitational field, the problem can be reduced to two restricted two-body systems for modeling purposes. To describe the motion, two coordinate systems are established, as illustrated in Fig. 1. The first is the inertial coordinate system $O_s - X_s Y_s Z_s$, where the solar center O_s serves as the origin. The X_s axis points towards the perihelion of the asteroid's orbit, the Y_s axis lies in the asteroid's orbital plane and is perpendicular to the X_s axis, and the Z_s axis satisfies the right-hand rule. The second system is the asteroid orbital coordinate system $O - XYZ$, established with the asteroid centroid O as the origin. The X axis extends along the asteroid's radius vector, while the Y axis lies in the asteroid's orbital plane, perpendicular to the X axis and pointing in the direction of the asteroid's motion (Liu et al., 2012). The asteroid moves counter clockwise along its orbit when viewed from the positive Z_s axis. The Z axis follows the right-hand rule.

In general, using Newton's laws of motion and the law of universal gravitation, the orbital dynamic equations of asteroid and probe in the inertial coordinate system can be derived as follows:

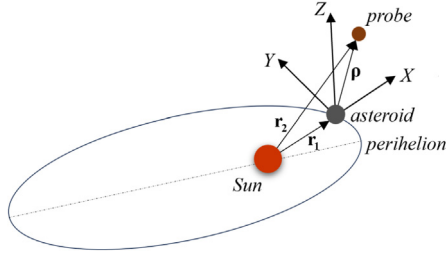


Fig. 1. Relative motion between the probe and the asteroid.

$$\ddot{\mathbf{r}}_1 = -\frac{\mu}{r_1^3} \mathbf{r}_1 + \mathbf{f}_1 \quad (15)$$

$$\ddot{\mathbf{r}}_2 = -\frac{\mu}{r_2^3} \mathbf{r}_2 + \mathbf{f}_2 \quad (16)$$

Here $\mu = GM$ is the heliocentric gravitational constant, G is the gravitational constant and M is the solar mass. The vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ represent the positions of the asteroid and the probe respectively. The scalars r_1 and r_2 denote their distances from the Sun to the centers of mass of the asteroid and the probe, while the vector $\mathbf{f}$ represents all the forces acting on a unit mass, excluding the heliocentric gravitational force. Accordingly, as illustrated in Fig. 1, the relative position vector of the probe with respect to the asteroid is defined as (Anton et al., 2017):

$$\boldsymbol{\rho} = \mathbf{r}_2 - \mathbf{r}_1 \quad (17)$$

By taking the second-order differentiation on Eq. (17) and substituting Eqs. (15) and (16), we obtain:

$$\ddot{\boldsymbol{\rho}} = \ddot{\mathbf{r}}_2 - \ddot{\mathbf{r}}_1 = -\frac{\mu}{r_2^3} \boldsymbol{\rho} + \mu \left(\frac{1}{r_1^3} - \frac{1}{r_2^3} \right) \mathbf{r}_1 + \Delta \mathbf{f} \quad (18)$$

Here,

$$\Delta \mathbf{f} = \mathbf{f}_2 - \mathbf{f}_1 = \mathbf{F}_a + \mathcal{F} \quad (19)$$

represents the resultant force between the probe and the asteroid, including all forces except the heliocentric gravitational attraction. In this equation, $\mathbf{F}_a$ denotes the gravitational attraction exerted by the asteroid, while $\mathcal{F}$ represents the model-free acceleration arising from uncertain disturbances and solar radiation pressure.

To express the above vectors in the asteroid-fixed coordinate frame $O-XYZ$, define $(\mathbf{e}_X \ \mathbf{e}_Y \ \mathbf{e}_Z)^T$ as the unit vectors of this frame. Consequently, we obtain:

$$\begin{cases} \boldsymbol{\rho} = x\mathbf{e}_X + y\mathbf{e}_Y + z\mathbf{e}_Z \\ \mathbf{r}_1 = r_1\mathbf{e}_X \\ \mathbf{r}_2 = \mathbf{r}_1 + \boldsymbol{\rho} = (r_1 + x)\mathbf{e}_X + y\mathbf{e}_Y + z\mathbf{e}_Z \\ \boldsymbol{\omega} = \omega\mathbf{e}_Z \\ \dot{\boldsymbol{\omega}} = \dot{\omega}\mathbf{e}_Z \\ \Delta \mathbf{f} = \Delta f_x \mathbf{e}_X + \Delta f_y \mathbf{e}_Y + \Delta f_z \mathbf{e}_Z \end{cases} \quad (20)$$

Subsequently, the scalar equations describing the relative motion of the probe in the asteroid's orbital coordinate system can be derived as follows:

$$\begin{cases} \ddot{x} = 2\omega\dot{y} + \omega^2x + \dot{\omega}y - \frac{\mu}{r_2^3}(x + r_1) + c r_1 + \Delta f_x \\ \ddot{y} = -2\omega\dot{x} + \left(\omega^2 - \frac{\mu}{r_2^3}\right)y - \dot{\omega}x + \Delta f_y \\ \ddot{z} = -\frac{\mu}{r_2^3}z + \Delta f_z \end{cases} \quad (21)$$

Here $r_1 = \frac{a(1-e^2)}{1+e\cos\theta}$ (Anton et al., 2017), where a denotes the semi-major axis, e the eccentricity, and θ the true anomaly.

The distance $r_2 = \sqrt{(r_1 + x)^2 + y^2 + z^2}$, while $\omega = \dot{\theta}$, $\dot{\omega} = -2ce\sin\theta$ represents the orbital angular velocity and orbital angular acceleration of the asteroid, respectively. By substituting $c = \frac{\mu}{r_1^3}$ into the expression for r_1 and applying $n = \frac{2\pi}{T} = \sqrt{\frac{\mu}{a^3}}$, can obtain:

$$c = n^2 \left(\frac{1 + e \cos f}{1 - e^2} \right)^3$$

Here, n denotes the mean angular velocity of the asteroid in the elliptical orbit.

Assume that the asteroid is located at the perihelion of its orbit at the initial time, i.e. $t_p = t_0 = 0$, then the mean anomaly can be expressed as $M = n(t - t_p) = nt$, and the true anomaly θ can be expanded in a series:

$$\begin{aligned} \theta = M + \left(2e - \frac{e^3}{4} \right) \sin M + \frac{5}{4}e^2 \sin 2M + \frac{13}{12}e^3 \\ \times \sin 3M + \dots \end{aligned} \quad (22)$$

By substituting the mean anomaly M into Eq. (22) and applying a second-order approximation, the expression for the true anomaly is obtained as:

$$\theta \approx nt + \left(2e - \frac{e^3}{4} \right) \sin(nt) + \frac{5}{4}e^2 \sin 2(nt) \quad (23)$$

Consequently, the orbital angular velocity ω can be expressed as:

$$\omega = \dot{\theta} = n + n \left(2e - \frac{e^3}{4} \right) \cos(nt) + \frac{5}{2}ne^2 \cos 2(nt) \quad (24)$$

The gravitational potential function of the asteroid can be derived using the gravitational field model based on its harmonic expansion (Xin, 2017):

$$U(\rho) = \frac{\mu_a}{\rho} \left\{ 1 + \left(\frac{R_0}{\rho} \right)^2 \left[\frac{1}{2} C_{20} \left(3 \frac{z^2}{\rho^2} - 1 \right) + 3 C_{22} \frac{x^2 - y^2}{\rho^2} \right] \right\} \quad (25)$$

Here μ_a denotes the gravitational constant of the asteroid, and R_0 is its reference radius. The coefficients C_{20} , C_{22} represent the second-degree spherical harmonic terms in the asteroid's gravitational potential expansion. The gravitational force is obtained by taking the gradient of the gravitational potential function:

$$\mathbf{F}_a = -\mathbf{grad}[U(\rho)] \quad (26)$$

Considering that the distance r_1 (between the asteroid and the Sun) and r_2 (between the probe and the Sun) are both much greater than the distance ρ between the probe and the asteroid, the scalar equation set (21) governing the relative motion in the asteroid's orbital coordinate system can be simplified as follows:

$$\begin{cases} \ddot{x} = 2\omega\dot{y} + (\omega^2 + 2c)x + \dot{\omega}y + \Delta f_x \\ \ddot{y} = -2\omega\dot{x} - \dot{\omega}x + \Delta f_y \\ \ddot{z} = -\omega^2 z + \Delta f_z \end{cases} \quad (27)$$

where $\Delta f_x, \Delta f_y, \Delta f_z$ denote the perturbation forces along the corresponding axis. If gravitational perturbations and other uncertainties are neglected ($\Delta f = 0$), Eq. (27) can be further simplified as:

$$\begin{cases} \ddot{x} = 2\omega\dot{y} + (\omega^2 + 2c)x + \dot{\omega}y \\ \ddot{y} = -2\omega\dot{x} - \dot{\omega}x \\ \ddot{z} = -\omega^2 z \end{cases} \quad (28)$$

By comparing Eq. (28) with Eq. (9), it is evident that the equation satisfies the self-adjoint condition defined in Eq. (11). Consequently, the Lagrangian function of the asteroid-probe dynamic system can be formulated:

$$L = \frac{1}{2}\dot{x}^2 + \frac{1}{2}\dot{y}^2 + \frac{1}{2}\dot{z}^2 - \omega y\dot{x} + \omega x\dot{y} + \frac{1}{2}(\omega^2 + 2c)x^2 - \frac{1}{2}\omega^2 z^2 \quad (29)$$

By applying the discrete Hamilton's principle, the discrete Lagrangian function of the system can be obtained, which can subsequently be used to perform numerical simulations through the variational Integrators.

4. Numerical simulation

4.1. Simulation setup

In this section, Eros is selected as the simulation example. The simulations are performed on a personal computer equipped with an AMD Ryzen AI 9 2.00 GHz CPU and 32 GB LPDDR5X memory. The first Near-Earth Asteroid discovered, Eros (found in 1898), is chosen for the following simulation analysis. Eros is an irregularly shaped Amor-type asteroid. Its orbital motion exhibits higher stability when the orbital radius is approximately 31 km. Given that the mass of the Sun is M , the mass of Eros is m , the gravitational constant is G , which is an exact (infinitely precise) constant. The heliocentric gravitational constant $\mu = GM$, the gravitational constant of Eros is $\mu_a = Gm$, the values of each orbital element are given as follows: eccentricity $e = 0.222574$, semi-major axis $a = 1.45793\text{AU}$, reference radius $R_0 = 16\text{km}$, the spherical harmonic coefficients in the gravitational potential function are $C_{20} = -0.03$, $C_{22} = 0.38$. The initial position and velocity of the probe in its relative motion with respect to Eros are: $\rho_0 = [23\text{km} \ 21\text{km} \ 0\text{km}]^T$, $\dot{\rho}_0 = [1.5 \times 10^{-3}\text{km/s} \ -0.8 \times 10^{-3}\text{km/s} \ 0\text{km/s}]^T$ respectively.

4.2. Simulation without gravitational perturbation

To simplify the computational process, the Lagrangian function expressed in Eq. (29) is discretized using the second-order Störmer-Verlet method:

$$\begin{aligned} L_d = & \frac{1}{2} \frac{(x_{k+1} - x_k)^2}{h} + \frac{1}{2} \frac{(y_{k+1} - y_k)^2}{h} + \frac{1}{2} \frac{(z_{k+1} - z_k)^2}{h} \\ & - \frac{\omega}{2} (y_{k+1} + y_k)(x_{k+1} - x_k) \\ & + \frac{\omega}{2} (x_{k+1} + x_k)(y_{k+1} - y_k) + \frac{h}{8} (\omega^2 + 2c)(x_{k+1} + x_k)^2 \\ & - \frac{h}{8} \omega^2 (z_{k+1} + z_k)^2 \end{aligned} \quad (30)$$

by substituting the discrete Lagrangian function given in Eq. (30) into Eq. (5), the discrete Hamiltonian equations of the system-under the condition that gravitational perturbations are neglected-can be derived as follows:

$$\begin{cases} p_{k,x} = \frac{x_{k+1} - x_k}{h} - \omega y_{k+1} - \frac{h}{4} (\omega^2 + 2c)(x_{k+1} + x_k) \\ p_{k,y} = \frac{y_{k+1} - y_k}{h} + \omega x_{k+1} \\ p_{k,z} = \frac{z_{k+1} - z_k}{h} + \frac{h}{4} \omega^2 (z_{k+1} + z_k) \\ p_{k+1,x} = \frac{x_{k+1} - x_k}{h} - \omega y_k + \frac{h}{4} (\omega^2 + 2c)(x_{k+1} + x_k) \\ p_{k+1,y} = \frac{y_{k+1} - y_k}{h} + \omega x_k \\ p_{k+1,z} = \frac{z_{k+1} - z_k}{h} - \frac{h}{4} \omega^2 (z_{k+1} + z_k) \end{cases} \quad (31)$$

In the absence of gravitational perturbations from the asteroid, this study employs the fourth order Runge-Kutta method (RK4) and the eighth order Runge-Kutta method (RK8), together with variational integrators (VI), to analyze numerically the relative motion between the asteroid and the probe over periods of 30 days (Fig. 2, time step 100 s) and 2000 days (Fig. 3, time step 2000 s). The Runge-Kutta method is a classic and widely used numerical integration scheme, noted for its straightforward implementation and explicit integration property. In this work, both RK4 and RK8 are implemented with fixed time steps: RK4 serves as a baseline traditional integrator, while RK8 provides a high order counterpart for evaluating long term simulation performance.

As can be observed from Figs. 2 and 3, the results obtained using the RK4 method and RK8 method are essentially consistent with those derived from the variational integrators:

The subsequent analysis calculates the temporal variation of the system's energy error during the relative motion, as illustrated in Fig. 4:

Fig. 4 illustrates the temporal variation of the system's energy error during the relative motion simulation, comparing the fourth-order Runge-Kutta method (RK4), the eighth-order Runge-Kutta method (RK8), and the variational integrators (VI).

The left one corresponds to a simulation time span of 1 day with a time step of $h = 100\text{s}$. As shown in the Fig. 4, the energy error obtained by RK4 and RK8 almost overlap and gradually increase with time, indicating a slight loss of energy conservation even over a short

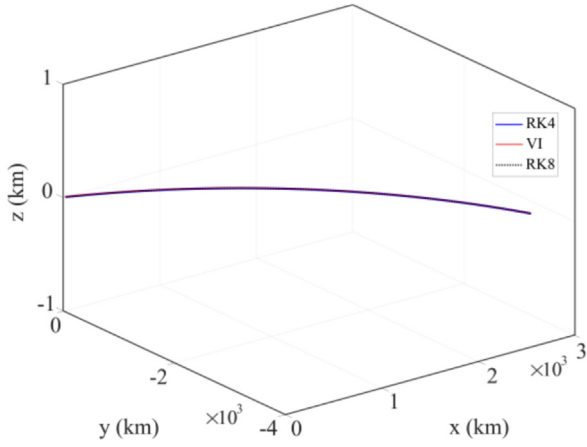


Fig. 2. 30-day probe motion trajectory without gravitational perturbation.

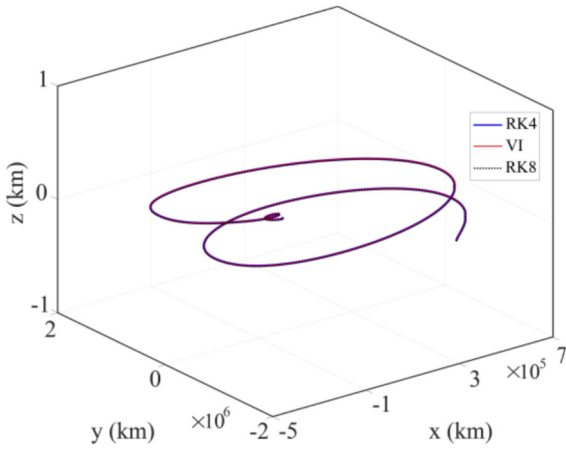


Fig. 3. 2000-day probe motion trajectory without gravitational perturbation.

integration period. In contrast, the variational integrators maintain an energy error that is nearly zero throughout the simulation, demonstrating excellent short-term energy preservation performance. The right one corresponds to a

simulation time span of 2000 days with a time step of $h = 2000s$. Over this long-term integration, both RK4 and RK8 exhibit significant oscillations and growth in energy error, with amplitudes roughly one order of magnitude larger than those of the variational integrators. The variational integrators, on the other hand, maintain a consistently small and stable energy error, confirming its superior long-term energy conservation capability because it's derived from the discrete Hamilton' principle.

Given that the energy-error curves of RK4 and RK8 are almost completely overlapping, a more comprehensive comparison of how the relative energy errors of the three algorithms evolve over different simulation durations was carried out, with the corresponding result was presented in Fig. 5:

As shown in Fig. 5, when the simulation time is short ($10^5 s$, $10^6 s$), the energy error of the Runge-Kutta methods (RK4 and RK8) remains extremely small—even smaller than that of the variational integrators (VI). This is attributed to the Runge-Kutta methods' high-order accuracy and explicit integration properties, which enable effective preservation of the system's energy in short-term simulations with no significant error accumulation. As time evolves to $10^7 s$, the three curves gradually approach and intersect—this occurs because the error of the Runge-Kutta methods have started to accumulate, while that of the variational integrators remains stable. However, upon extending the simulation time to $10^8 s$ and $10^9 s$, the energy error of the Runge-Kutta methods grows rapidly, far exceeding that of the variational integrators. This result further demonstrates that the variational integrators can maintain the system's symplectic structure even in long-term integration, thereby effectively suppressing energy drift.

4.3. Simulation with gravitational perturbation

When considering the influence of the asteroid's gravitational perturbation on the probe's trajectory, the Lagran-

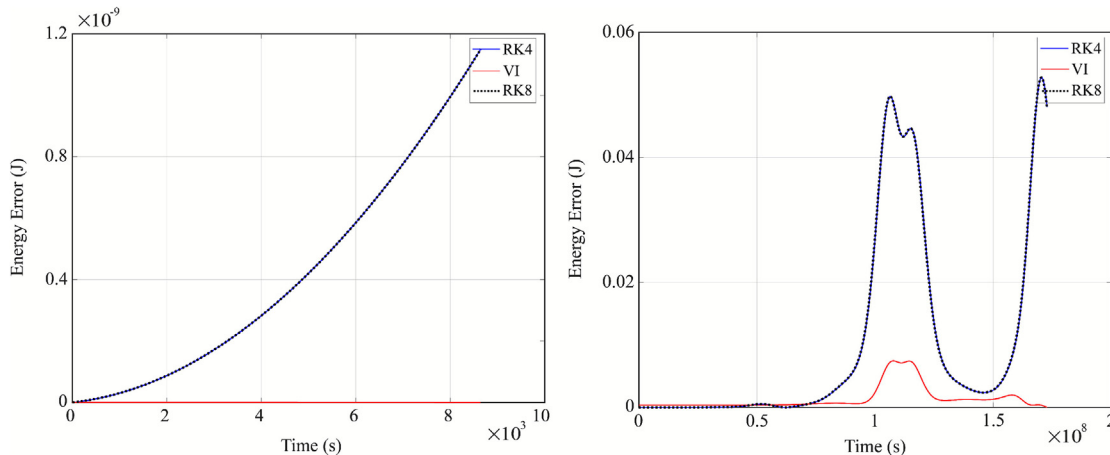


Fig. 4. Energy error over time without gravitational perturbation.

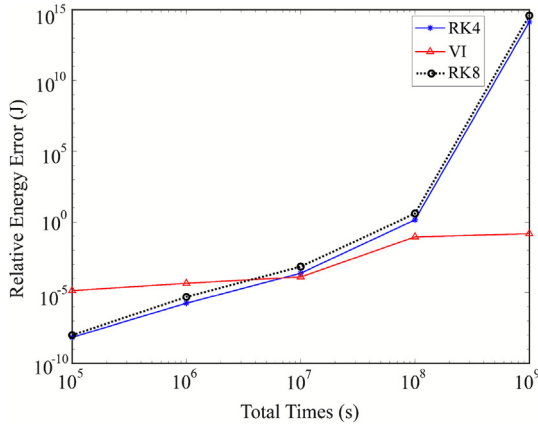


Fig. 5. Relative energy error over total times without gravitational perturbation.

gian function for the relative motion of the system can be expressed as:

$$L' = L - U(\rho) \quad (32)$$

The second-order Störmer-Verlet method is employed to discretize Eq. (32). Consequently, the discrete Lagrangian function of the probe system, accounting for the gravitational perturbations of the asteroid, can be derived as follows:

$$\begin{aligned} L'_d = L_d - \frac{h}{2} \frac{\mu_a}{(x_k^2 + y_k^2 + z_k^2)^{12}} - \frac{3h}{4} \frac{\mu_a R_0^2 C_{20} z_k^2}{(x_k^2 + y_k^2 + z_k^2)^{52}} + \frac{h}{4} \frac{\mu_a R_0^2 C_{20}}{(x_k^2 + y_k^2 + z_k^2)^{32}} \\ - \frac{3h}{2} \frac{\mu_a R_0^2 C_{22} (x_k^2 - y_k^2)}{(x_k^2 + y_k^2 + z_k^2)^{52}} - \frac{h}{2} \frac{\mu_a}{(x_{k+1}^2 + y_{k+1}^2 + z_{k+1}^2)^{12}} - \frac{3h}{4} \frac{\mu_a R_0^2 C_{20} z_{k+1}^2}{(x_{k+1}^2 + y_{k+1}^2 + z_{k+1}^2)^{52}} \\ + \frac{h}{4} \frac{\mu_a R_0^2 C_{20}}{(x_{k+1}^2 + y_{k+1}^2 + z_{k+1}^2)^{32}} - \frac{3h}{2} \frac{\mu_a R_0^2 C_{22} (x_{k+1}^2 - y_{k+1}^2)}{(x_{k+1}^2 + y_{k+1}^2 + z_{k+1}^2)^{52}} \end{aligned} \quad (33)$$

By applying Eq. (31) and substituting Eq. (33) into Eq. (5), the discrete Hamiltonian equations for the probe system, which incorporate the gravitational perturbations of the asteroid, can be obtained:

$$\begin{cases} p'_{k,x} = p_{k,x} - A_k x_k - 5B_k x_k z_k^2 + C_k x_k + D_k (7x_k y_k^2 - 3x_k^3 + 2x_k z_k^2) \\ p'_{k,y} = p_{k,y} - A_k y_k - 5B_k y_k z_k^2 + C_k y_k + D_k (3y_k^3 - 7y_k x_k^2 - 2y_k z_k^2) \\ p'_{k,z} = p_{k,z} - A_k z_k + B_k (2z_k x_k^2 + 2z_k y_k^2 - 3z_k^3) + C_k z_k - 5D_k z_k (x_k^2 - y_k^2) \\ p'_{k+1,x} = p_{k+1,x} + A_{k+1} x_{k+1} + 5B_{k+1} x_{k+1} z_{k+1}^2 - C_{k+1} x_{k+1} - D_{k+1} (7x_{k+1} y_{k+1}^2 - 3x_{k+1}^3 + 2x_{k+1} z_{k+1}^2) \\ p'_{k+1,y} = p_{k+1,y} + A_{k+1} y_{k+1} + 5B_{k+1} y_{k+1} z_{k+1}^2 - C_{k+1} y_{k+1} - D_{k+1} (3y_{k+1}^3 - 7y_{k+1} x_{k+1}^2 - 2y_{k+1} z_{k+1}^2) \\ p'_{k+1,z} = p_{k+1,z} + A_{k+1} z_{k+1} - B_{k+1} (2z_{k+1} x_{k+1}^2 + 2z_{k+1} y_{k+1}^2 - 3z_{k+1}^3) - C_{k+1} z_{k+1} \\ + 5D_{k+1} z_{k+1} (x_{k+1}^2 - y_{k+1}^2) \end{cases} \quad (34)$$

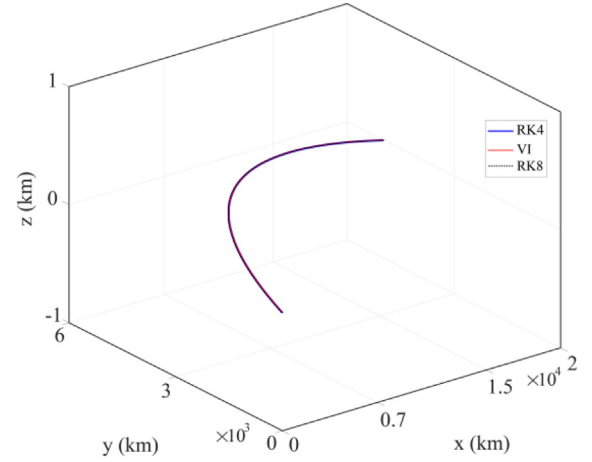


Fig. 6. 30-day probe motion trajectory with gravitational perturbation.

The coefficients in Eq. (34) are defined as follows:

$$\begin{cases} A_k = \frac{h}{2} \frac{\mu_a}{(x_k^2 + y_k^2 + z_k^2)^{3/2}}, B_k = \frac{3h}{4} \frac{\mu_a R_0^2 C_{20} x_k}{(x_k^2 + y_k^2 + z_k^2)^{7/2}} \\ C_k = \frac{3h}{4} \frac{\mu_a R_0^2 C_{20}}{(x_k^2 + y_k^2 + z_k^2)^{5/2}}, D_k = \frac{3h}{2} \frac{\mu_a R_0^2 C_{22}}{(x_k^2 + y_k^2 + z_k^2)^{7/2}} \\ A_{k+1} = \frac{h}{2} \frac{\mu_a}{(x_{k+1}^2 + y_{k+1}^2 + z_{k+1}^2)^{3/2}}, B_{k+1} = \frac{3h}{4} \frac{\mu_a R_0^2 C_{20} x_{k+1}}{(x_{k+1}^2 + y_{k+1}^2 + z_{k+1}^2)^{7/2}} \\ C_{k+1} = \frac{3h}{4} \frac{\mu_a R_0^2 C_{20}}{(x_{k+1}^2 + y_{k+1}^2 + z_{k+1}^2)^{5/2}}, D_{k+1} = \frac{3h}{2} \frac{\mu_a R_0^2 C_{22}}{(x_{k+1}^2 + y_{k+1}^2 + z_{k+1}^2)^{7/2}} \end{cases}$$

Using the same parameters as previously described, numerical simulations of the asteroid-probe relative motion over a 30-day period were carried out using the fourth-order Runge-Kutta method, the eighth-order Runge-Kutta, and the variational integrators. The corresponding relative motion trajectories of the asteroid and the probe are illustrated in Fig. 6, where the simulation was performed with a time step of $h = 100$ s. Furthermore, when the simulation time was extended to 2000 days, with a time step of $h = 2000$ s, the resulting relative motion trajectories are shown in Fig. 7.

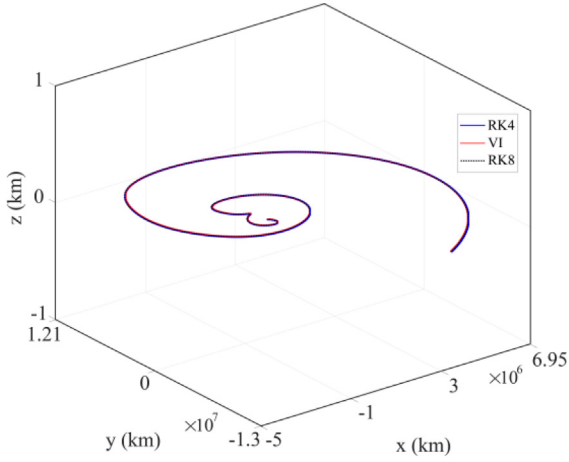


Fig. 7. 2000-day probe motion trajectory with gravitational perturbation.

The temporal variation of the energy error within the asteroid-probe system, considering the gravitational perturbations of the asteroid is computed and illustrated in Fig. 8:

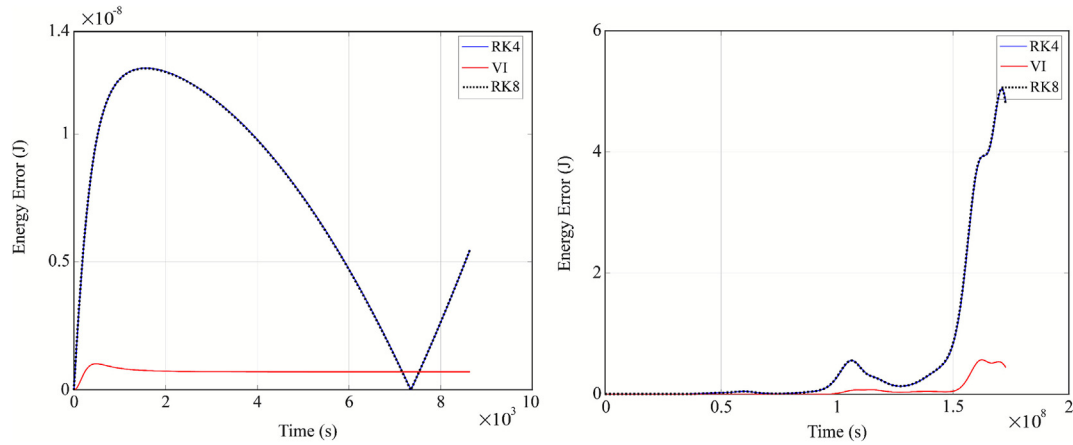


Fig. 8. Energy error over time with gravitational perturbation.

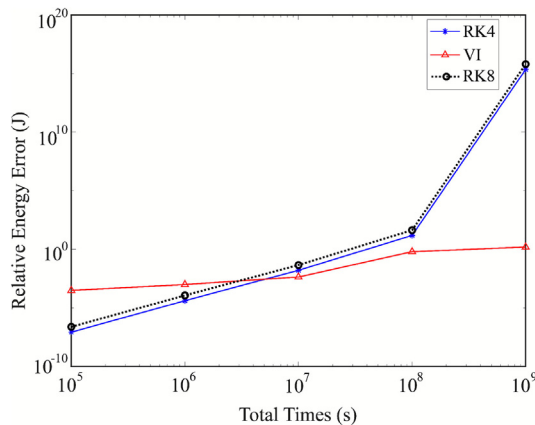


Fig. 9. Relative energy error over total times with gravitational perturbation.

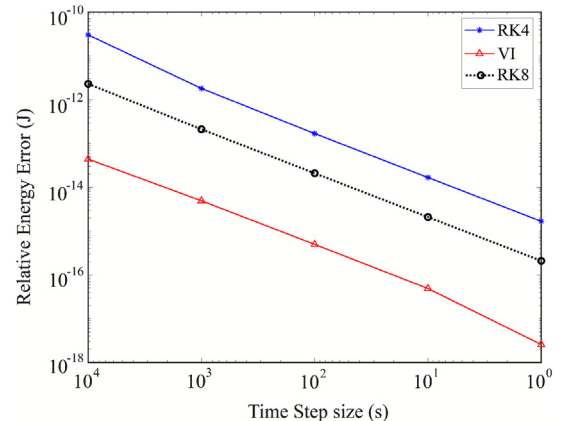


Fig. 10. Convergence analysis of the energy error over time step size.

Table 1
Computation time comparison of three methods.

Three Methods	Total Times (s)		
	10^5	10^6	10^7
RK8	0.1	0.17	0.63
RK4	0.04	0.06	0.18
VI	0.01	0.02	0.24

relative energy error that accumulates over the entire simulation duration. The results are presented in Fig. 10, where the variation in this accumulative error is plotted for the three algorithms under the same total time but with different time step sizes.

The slope of each curve reflects the convergence rate of the corresponding numerical integration methods. As can be observed, all three methods-RK4, RK8, and VI exhibit approximately linear convergence in the logarithmic scale, indicating consistent numerical stability. However, the slope of the VI curve is steeper, suggesting that the variational integrators achieve faster convergence with decreasing time step size. In addition, the overall magnitude of the energy error for the VI method is significantly smaller than those of RK4 and RK8 across all time step sizes. This indicates that the VI method maintains superior energy preservation properties, even when larger time steps are used.

Finally, the difference in total computation time was compared among these three methods, as shown in Table 1:

The data indicates that over short durations, the computational efficiency of VI surpasses that of the Runge–Kutta method. However, when extending the time, the computational efficiency of VI becomes slightly lower than that of RK4 but remains higher than RK8, owing to the cost of its implicit integration scheme relative to RK4, but the even greater per-step computational expense of the high-order RK8 method.

Through a comparative analysis between the scenarios without gravitational perturbations (depicted in Figs. 2 and 3) and those incorporating gravitational perturbations (shown in Figs. 6 and 7), it becomes evident that the gravitational disturbances induced by the asteroid itself play a crucial role in asteroid exploration. Specifically, under the influence of asteroid-induced gravitational perturbations, the deviation in the probe's motion trajectory increases by nearly two orders of magnitude. This finding indicates that gravitational perturbations from asteroids constitute a critical factor that must be carefully considered during probe orbiting and maneuvering operations around asteroids.

5. Conclusion

This research constructs the orbital dynamics model of an asteroid probe by integrating the law of universal gravitation, Newton's Second Law, and the spherical harmonic gravitational field model of asteroids. Then, based on the

inverse problem of Lagrangian mechanics, the Lagrangian function of the orbital dynamics model is derived. The discrete Hamilton's principle is further applied to discretize the model, resulting in the discrete Hamiltonian equations. Next, taking the 433rd asteroid Eros as a case, the variational integrators are used to perform simulation analyses with and without the asteroid's gravitational perturbation and finding that regardless of whether the gravitational perturbation of the asteroid is considered, the variational integrators (using Störmer-Verlet method) can more stably maintain the system energy in long-term simulations compared with the traditional RK4 and RK8. By comparing the simulations with and without the gravitational perturbation, it turns out that is a non-negligible key factor in the orbiting process. This work provides a high-precision numerical calculation method for the orbital optimization design of asteroid orbiting probes, demonstrating the superiority of variational integrators in studying the orbital dynamics of asteroid probes. Future research will primarily focus on integrating discrete mechanics with optimal control to explore trajectory optimization suitable for asteroid probes, achieving real-time optimal design of flyby orbits and landing operations for deep-space probes.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Shixing Liu Yongxin Guo reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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